

Persulfidation and phosphorylation of transcription factor SlWRKY6 differentially regulate tomato fruit ripening

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Abstract

Cysteine desulfhydrase catalyses the generation of the signaling molecule hydrogen sulfide (H₂S) in plants. In this study, we found that H₂S can inhibit tomato (*Solanum lycopersicum*) fruit ripening and SlWRKY6 undergoes differential protein persulfidation in SlLCD1-overexpressing leaves. Then, further study indicated that SlWRKY6 could be persulfidated by H₂S at Cys396. By construction of *slwrky6* mutants and SlWRKY6-OE lines, we found that SlWRKY6 positively regulates leaf senescence and fruit ripening by activating the transcription of ripening-related genes STAYGREEN 1 (SlSGR1) and Senescence-Associated Gene 12 (SlSAG12). In addition, SlWRKY6 interacted with kinase SlMAPK4 and was phosphorylated at Ser33. Dual-luciferase transient expression assays and electrophoretic mobility shift assays indicated that SlWRKY6 persulfidation attenuated its transcriptional regulation of target genes SlSGR1 and SlSAG12, whereas SlWRKY6 phosphorylation by SlMAPK4 activated the transcription of target genes to promote fruit ripening. Moreover, we provided evidence that SlWRKY6 persulfidation attenuated its SlMAPK4-mediated phosphorylation to inhibit tomato fruit ripening. By transient expression of SlWRKY6, SlWRKY6^{C396A}, SlWRKY6^{S33A}, and SlWRKY6^{S33D} in *slwrky6* fruits, we found that SlWRKY6 persulfidation attenuated the expression of SlSGR1 and SlSAG12 thereby delaying tomato fruit ripening, while SlWRKY6 phosphorylation increased the expression of target genes. As tomato fruits ripened, endogenous H₂S production decreased, while SlMAPK4 expression increased. Therefore, our findings reveal a model in which SlWRKY6 persulfidation due to higher endogenous H₂S levels in un-ripened fruit inhibits its ability to activate SlSGR1 and SlSAG12 expression, while SlWRKY6 phosphorylation by SlMAPK4 activates its transcriptional activity, thereby promoting tomato fruit ripening.

Introduction

Tomato (*Solanum lycopersicum*) is a fruit and vegetable crop rich in nutrients that is widely grown in the world. Due to its clear genetic background and short growth cycle, it is often used as a model plant for studying the mechanism of fruit ripening and senescence (Wu et al. 2020). Fruit ripening is a complex developmental process involving a variety of physiological and biochemical reactions such as changes in fruit color, texture, and accumulation of soluble sugars, carotenoids, and volatile organic compounds (Klee and Giovannoni 2011; Giovannoni et al. 2017). The study of fruit ripening and senescence is crucial to reveal the mechanism of fruit development, which can help improve yield, optimize quality, and improve postharvest storage of tomatoes.

Transcription factors (TFs) play essential roles during plant growth and development by regulating the transcription of downstream genes. In recent years, a plethora of TFs have been reported to play important regulatory roles in tomato fruit ripening. For instance, MADS-box family protein RIPENING INHIBITOR, NON-RIPENING, and COLORLESS NON-RIPENING have been identified as positive regulators of fruit ripening (Karlova et al. 2014). In addition, WRKY TFs are an important family of TF that participates in a variety of biotic, abiotic, or physiological responses (Eulgem and Somssich 2007). WRKY TFs are zinc finger-type proteins containing WRKYGQK heptapeptide at the N-terminus and zinc finger structure at the C-terminus (C₄-CX₂₂₋₂₃HXH or

CX₂CX₂₃HXC) and specifically recognize cis-acting element called the W-box (TTGACT/C) in the promoters of target genes (Rushton et al. 2010). Accumulating studies have shown that WRKY TFs are involved in plant growth and development. For instance, wheat (*Triticum aestivum*) TaWRKY51 promotes lateral root formation by binding to the W-box in the promoter and blocking the expression of ACS genes involved in ethylene biosynthesis (Hu et al. 2018). MdWRKY17 can directly promote the expression of the salicylic acid degradation gene Downy Mildew Resistant 6 (MdDMR6), resulting in reduced resistance to *Colletotrichum fructicola* (Shan et al. 2021). SlWRKY35 could directly activate the expression of SlDXS1 leading to the accumulation of carotenoids and thus promoting tomato fruit ripening (Yuan et al. 2022).

Hydrogen sulfide (H₂S) is a gaseous signaling molecule that can effectively regulate growth and developmental processes such as seed germination, fruit ripening, senescence, and biotic and abiotic stress responses in plants (Zhang et al. 2008; Hu et al. 2020). H₂S could substantially inhibit the ripening and senescence of postharvest fruits and prolong their storage time as found in strawberry, tomato, and pear fruits (Hu et al. 2012; Ziogas et al. 2018; Yao et al. 2020). The mechanism of H₂S in delaying fruit ripening and senescence may involve the regulation of redox balance, inhibiting ethylene biosynthesis and antagonizing ethylene effect and signaling pathway and energy metabolism in fruits and the integrity of cell wall

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and cell membrane (Jia et al. 2018; Hu et al. 2020; Yao et al. 2020).

As a signaling molecule, H₂S is believed to accomplish its function in vivo through protein persulfidation (Filipovic et al. 2018). As a recently discovered type of post-translational modification of proteins, protein S-sulfhydration modification, also known as persulfidation, refers to the transformation of a sulfhydryl group (–SH) on a cysteine (Cys) residue of a protein into a persulfidated state (–SSH). Aroca et al. (2017) identified 2015 proteins with sulfhydryl modifications in *Arabidopsis thaliana*, which are involved in multiple processes of plant growth and development, suggesting that protein sulfhydryl modifications may be a widespread form of post-translational modification of proteins. Recent studies have found that the autophagy-related protein ATG18 can undergo persulfidation at the Cys103 site in response to H₂S and thus inhibit the autophagy process in *A. thaliana* under nutrient deprivation (Aroca et al. 2021). During H₂S-induced stomatal closure, SnRK2.6 can respond to H₂S signal and undergo persulfidation at the Cys131 and Cys137 sites, leading to an increase in the activity of the SnRK2.6 kinase and facilitating its binding to the ABA-responsive element-binding factor ABF2, which in turn induces stomatal closure (Chen et al. 2021). Recently, H₂S was found to persulfidate E3 ubiquitin-protein ligase 3 BRG3 and reduce its ubiquitination activity, which in turn led to subsequently delayed ripening of tomatoes (Sun et al. 2023).

In plants, endogenous H₂S is produced mainly during cysteine catabolism. The cysteine desulfhydrases LCD (L-cysteine desulfhydrase) or DCD (D-cysteine desulfhydrase) use L-Cys and D-Cys, respectively, as substrates to produce H₂S (Alvarez et al. 2010; Scuffi et al. 2014; Zhao et al. 2023). Previously, we reported the function of *SILCD1* and *SILCD2* as negative regulators of fruit ripening, further confirming the role of endogenous H₂S in delaying ripening and senescence (Hu et al. 2020; Zhao et al. 2023). However, whether differential endogenous H₂S levels could cause differential levels of protein persulfidation still needs further research.

In addition to protein persulfidation, phosphorylation has long been considered as an important way for regulating protein functions (Zhang et al. 2023). When cells sense signals through receptors, the signals are transmitted down the MAPKKK–MAPKK–MAPK cascade through phosphorylation, and the activated MAPKs regulate protein functions by phosphorylating downstream TFs, protein kinases, and other substrates to achieve signal transduction (Danquah et al. 2014; Wei et al. 2023). Recently, phosphorylated TFs modified by MAPK kinase have been extensively studied in plants. For instance, banana MabZIP21 (leucine zipper 21) phosphorylation mediated by MaMPK6-3 could increase the expression of softening-related genes, thereby promoting fruit ripening in banana (Wu et al. 2022). Phosphorylation of *SILBBX17* by *SILMPK1* and *SILMPK2* positively regulated cold tolerance in tomato (Song et al. 2023). Wei et al. (2023) showed that an ethylene-induced increase in MdMAPK3 kinase activity promoted the phosphorylation of MdNAC72 by MdMAPK3 and thus apple fruit softening. However, whether and how persulfidation interferes with phosphorylation in a single protein is still lack of evidence.

In the present study, differential persulfidated proteins in *SILCD1* overexpression leaves were analyzed by 4D-label-free LC/MS proteomics and it was noticed that a peptide in a WKRY protein (A0A3Q7F586) which was later named as SIWRKY6 showed higher intensity of persulfidation. By CRISPR/Cas9 and gene overexpression, we showed that SIWRKY6 was a positive regulator of leaf senescence and tomato fruit ripening by activating *STAYGREEN 1* (*SISGR1*) and *Senescence-Associated Gene 12* (*SISAG12*) transcription.

Further evidence indicated that persulfidation of SIWRKY6 by H₂S attenuated its transcriptional activity to target genes *SISGR1* and *SISAG12*. However, SIWRKY6 phosphorylation by *SILMAPK4* could activate the expression of *SISGR1* and *SISAG12* to promote tomato fruit ripening. Moreover, we provided evidence that SIWRKY6 persulfidation mediated by H₂S attenuated *SILMAPK4*-mediated phosphorylation of SIWRKY6 to inhibit tomato fruit ripening. Therefore, the current study provides a solid theoretical basis into the mechanism of persulfidation and phosphorylation of a single protein SIWRKY6 in regulating tomato fruit ripening.

Results

Identification of persulfidation modification at Cys396 of SIWRKY6 protein

To explore the role of H₂S in regulating tomato fruit ripening, green ripening stage tomato fruits were fumigated with H₂S. As shown in Supplementary Fig. S1, H₂S substantially delayed tomato fruit ripening, whether H₂S functioned through its mediation of target protein persulfidation is still unclear. Previously, we created a *SILCD1* overexpression tomato line which encoding an H₂S generating enzyme LCD catalyzing L-cysteine to give rise H₂S, ammonia, and pyruvate (Hu et al. 2021). *SILCD1* overexpression was found to delay dark-induced senescence and fruit ripening. Therefore, we carried out 4D-label-free LC/MS proteomics to investigate potential differentially persulfidated proteins in *SILCD1* overexpression and wild-type (WT) leaves. Among the 119 peptides that underwent persulfidation, it was noticed that a peptide in a WKRY protein (A0A3Q7F586, encoding by Solyc02g080890.2) showed a higher intensity of persulfidation. By homologous comparison, the WKRY protein showed high sequence similarity to WKRY6 in *A. thaliana* and therefore named SIWRKY6 in tomato. To investigate whether the SIWRKY6 protein could undergo persulfidation in response to H₂S, in vitro persulfidation reaction was performed using H₂S donor NaHS as the sulfhydryl agent, followed by detection by modified biotin-switch method. As shown in Fig. 1A, NaHS treatment of purified HIS–SIWRKY6 protein induced the occurrence of persulfidation and the persulfidation signal increased as NaHS concentration increased. When dithiothreitol (DTT) was added, the signal of persulfidation was eliminated. The result indicated that the recombinant HIS–SIWRKY6 protein expressed in *Escherichia coli* could be sulfhydryl modified by H₂S and the sulfhydryl group could be reduced by DTT reductant. To determine the amino acid sites in SIWRKY6 that were persulfidated, purified HIS–SIWRKY6 protein was treated with exogenous H₂S and then analyzed by LC-MS/MS. The data revealed that the Cys site at Position 396 of SIWRKY6 protein could undergo persulfidation (Fig. 1B and Supplementary Data Set 1).

To further verify the persulfidated site, we mutated Cys396 to Ala in SIWRKY6 (Fig. 1C) and compared the effect of H₂S on the modification of SIWRKY6. It was found when Cys396 was mutated to Ala, persulfidation signal of SIWRKY6^{C396A} mutant protein was almost fully abolished shown by Western blotting results (Fig. 1D). These results suggested that SIWRKY6 could be modified by H₂S and the key site of persulfidation happened on Cys396.

SIWRKY6 is a positive regulator for fruit ripening of tomato

To investigate the function of SIWRKY6 in regulating leaf senescence or tomato fruit ripening, 2 lines of *slurky6* loss-function mutants by CRISPR/Cas9-mediated gene editing (*slurky6*#4, #11) and 2 lines of SIWRKY6 overexpressing plants SIWRKY6-FLAG

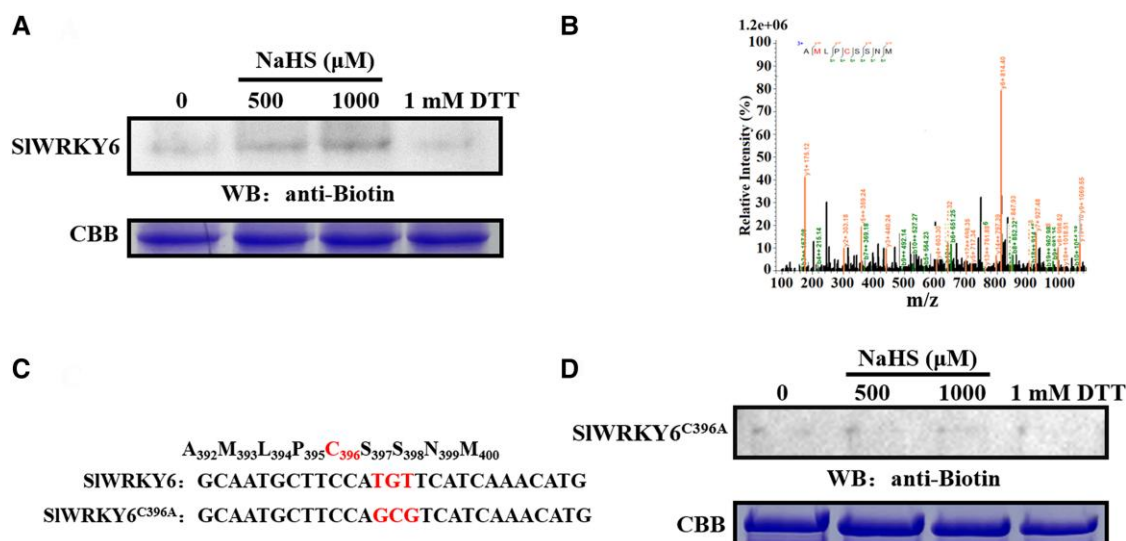


Figure 1. Identification of persulfidation site of SIWRKY6 protein. **A)** Detection of persulfidation of SIWRKY6 recombinant protein by modified biotin-switch. Purified SIWRKY6 protein was treated with different concentrations of H₂S donor NaHS (500 or 1,000 μM) or the reducing agent DTT (1 mM) for 1 h, respectively. Consistent protein loading as determined by Coomassie blue. **B)** The persulfidation site of the SIWRKY6 protein was detected by MS analysis. LC-MS/MS spectra show that AMLPC₃₉₆SSNM is a key site of SIWRKY6 persulfidation. The orange and green colors represent the y-ion and b-ion, respectively. **C)** Schematic diagram of the SIWRKY6 mutant protein (Cys₃₉₆-Ala, i.e. C396A). The red letters “TGT” to “GCG” represent the mutation of the codon of Cys to Ala. **D)** Detection of persulfidation state of SIWRKY6^{C396A} mutant protein by modified biotin-switch. CBB: Coomassie blue staining.

(SIWRKY6#1, #2) were generated. By genomic sequencing, it was found that *slwrky6#4* mutant had a homozygous 64 bp deletion which destroyed the gRNA1 on SIWRKY6, and the *slwrky6#11* mutant showed a homozygous 1 bp deletion adjacent to gRNA2 which caused frame-shift mutation to SIWRKY6 (Fig. 2, A and B). Specifically, *slwrky6#4* was mutated in the peptide sequence from the 28th amino acid residue that caused stopped translation at the 44th amino acid, and the translation of *slwrky6#11* stopped at the 235th amino acid, and both of the mutations destroyed the functional domain of SIWRKY6. SIWRKY6 overexpressing plants were confirmed by RT-qPCR and Western blotting, and the data indicated that SIWRKY6 expression level was approximately 100- and 400-fold higher in the SIWRKY6 overexpressing plants than WT, and the presence of SIWRKY6-FLAG was confirmed by immunoblotting with anti-FLAG antibodies, while no signal was found in WT and *slwrky6#4*, #11 (Fig. 2, C and D).

During the growth of SIWRKY6 plants, we found that SIWRKY6 overexpressing leaves senesced earlier compared to WT especially at 120 d of growth (Supplementary Figs. S2A and S3A). Therefore, we speculated that SIWRKY6 may promote leaf senescence. To clarify the regulatory mechanism of SIWRKY6, transcriptome analysis was performed in the leaves of WT and SIWRKY6 overexpression plants grown for 60 d. We identified 1,292 (675 upregulated and 617 downregulated) differentially expressed genes (DEGs) in SIWRKY6 compared to WT (Supplementary Fig. S2, B and C, Supplementary Data Set 2 and 3). Kyoto Encyclopedia of Genes and Genomes (KEGG) analyses revealed that upregulated genes were mainly involved in MAPK pathway (Supplementary Data Set 4), phenylpropanoid biosynthesis, and plant hormone signal transduction, and downregulated genes were preferentially associated with photosynthesis-antenna proteins and steroid biosynthesis (Supplementary Fig. S2, D and E). As shown in Supplementary Fig. S3A, SIWRKY6 overexpression leaves senesced earlier, while the leaves of *slwrky6* mutant were greener compared with WT, especially at 120 d of growth. The contents of total chlorophyll (including Chlorophyll a and Chlorophyll b) and carotenoids were consistent with the phenotype (Supplementary Fig.

S3, B to E). Heat map analysis of the DEGs associated with leaf senescence based on transcriptome data (Supplementary Data Set 5) showed that overexpression of SIWRKY6 significantly upregulated the expression of the chlorophyll degradation-related gene *SISGR1* and the senescence-related gene *SISAG12* (Supplementary Fig. S3F). The reliability of the RNA-seq data was verified by RT-qPCR detection, and the data of *SISGR1* and *SISAG12* expression in SIWRKY6 leaves matched with the expression pattern of transcriptome data (Supplementary Fig. S3, G and H). As *SISGR1* and *SISAG12* promoters both contained WRKY binding sites by analyzing cis-acting regulatory element on <https://bioinformatics.psb.ugent.be/webtools/plantcare/html/>, we speculated that they might be direct target genes of SIWRKY6. This result suggested that SIWRKY6 positively regulated tomato leave senescence possibly by regulating the expression of *SISGR1* and *SISAG12*. In addition, we analyzed the expression patterns of SIWRKY6, *SISGR1*, and *SISAG12* at an equatorial region in fruit tissues at 10, 20, and 30 d post-anthesis (DPA), mature green, breaker, and pink fruits by online tool on <https://tea.solgenomics.net/>. Generally, the data suggest that SIWRKY6, *SISGR1*, and *SISAG12* show increasing transcription during fruit ripening, implying that the genes might function in tomato fruit ripening (Supplementary Fig. S4).

Then, the role of SIWRKY6 in regulating tomato fruit ripening was investigated. As shown in Fig. 3, A and B, there was a clear difference between the WT, *slwrky6* and SIWRKY6 at fruit ripening. Fruits of SIWRKY6 overexpression plants turned red at 40 DPA, whereas WT fruits turned red at 45 DPA and later *slwrky6* mutant. By comparing the pigment content of WT, *slwrky6* and SIWRKY6 fruits, it was found that *slwrky6#4* and *slwrky6#11* fruits had higher chlorophyll content (Chlorophyll a and Chlorophyll b), while SIWRKY6#1 and SIWRKY6#2 fruits exhibited lower chlorophyll content compared to WT fruits during fruit ripening (Fig. 3, C to E). In addition, SIWRKY6#1 and SIWRKY6#2 fruits accumulated more carotenoids compared with 2 lines of mutants (*slwrky6#4* and *slwrky6#11*) fruits and WT fruits (Fig. 3F). As shown in RT-qPCR results of Fig. 3, G and H, SIWRKY6 significantly promoted the expression of *SISGR1* and *SISAG12* in tomato fruits.

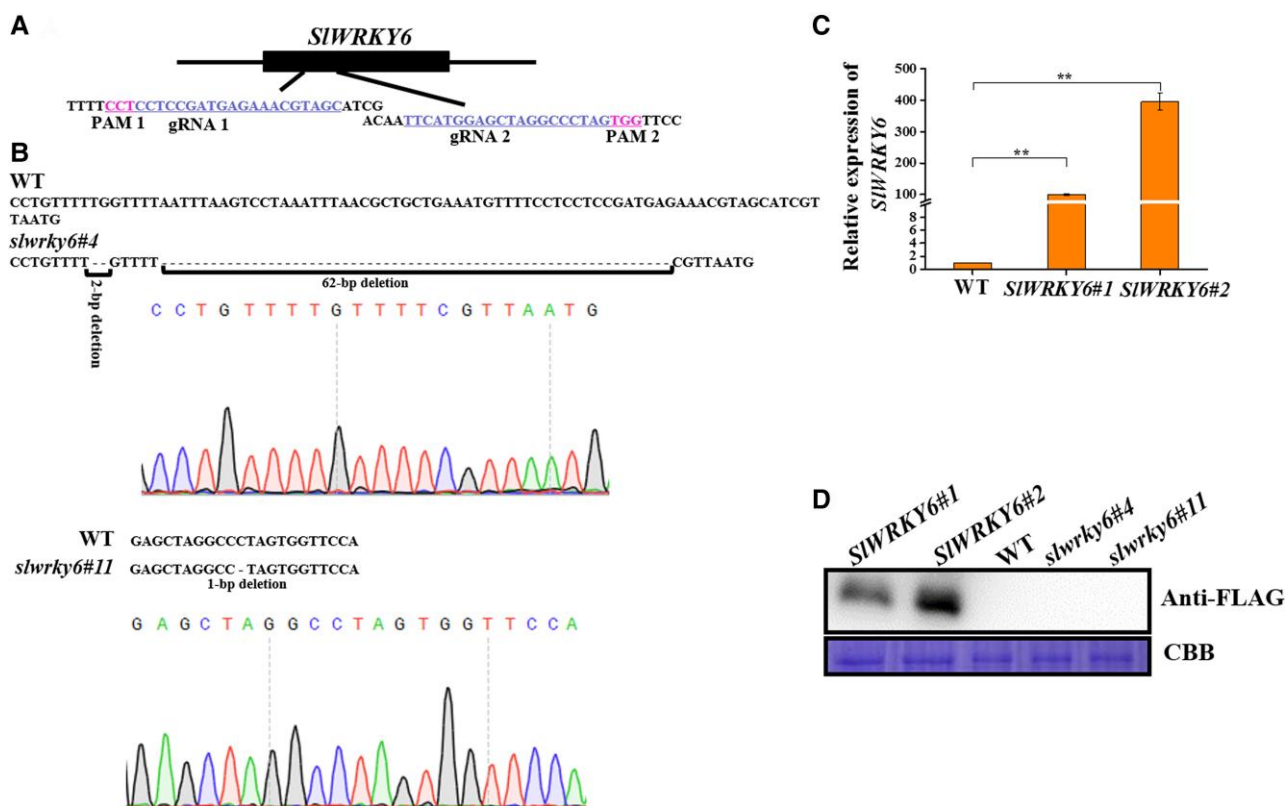


Figure 2. Construction of tomato *slwrky6* mutants through CRISPR/Cas9-mediated targeted mutagenesis and *SIWRKY6* overexpression lines. **A)** The schematic diagram shows the position and sequence of guide RNAs (gRNAs) in blue on the coding sequence of *SIWRKY6* and PAM (protospacer adjacent motif) in purple letters. **B)** Sequencing-based genotyping of mutations in *slwrky6#4* and *slwrky6#11*. The *slwrky6#4* mutant had a homozygous 64 bp deletion near gRNA1 and the *slwrky6#11* mutant had a homozygous 1 bp deletion near gRNA2 in *SIWRKY6*, respectively. WT, WT. **C)** Relative expression levels of *SIWRKY6* in WT and *SIWRKY6* overexpression (*SIWRKY6#1* and *SIWRKY6#2*) plants were determined by RT-qPCR. Data are shown as the means ($\pm$ standard deviation (SD)) of 3 biological replicates. The symbol ** indicates a significant difference compared to WT ($P < 0.01$, Student's t-test). **D)** Expression of *SIWRKY6*-FLAG protein in *SIWRKY6#1* and *SIWRKY6#2* was confirmed by immunoblot analysis using anti-FLAG with WT, *slwrky6#4* and *slwrky6#11* tomato plants as negative control. CBB: Coomassie blue staining after SDS-PAGE indicating of protein loading.

Expression of ethylene synthesis-related genes *SlACO1* and *SlACS2* were upregulated in the *SIWRKY6#1* and *SIWRKY6#2* fruits and downregulated in the *slwrky6#4* and *slwrky6#11* fruits compared with WT at 35, 40, and 45 DPA. Similar results were observed in the leaves (Supplementary Fig. S5). Then we analyzed the promoter sequence of *SlACO1* and *SlACS2*, and no W-box cis-elements (TTGACT/C) were found, suggesting that ethylene biosynthesis genes might not be the direct targets of *SIWRKY6* and the accelerated fruit ripening or leaf senescence might contribute to the changes in *SlACO1* and *SlACS2* expression. Therefore, all these data indicated that *SIWRKY6* is a positive regulator of leaf senescence and fruit ripening.

SIWRKY6 physically interacts with SIMAPK4

Then we screened potential phosphorylation sites of *SIWRKY6* on <http://gps.biocuckoo.cn/> and found that MAPK may target and phosphorylate *SIWRKY6* at the classical MAPK phosphorylating SP sites. Then, 5 mitogen-activated protein kinase genes (*SIMAPK1*, *SIMAPK2*, *SIMAPK3*, *SIMAPK4*, and *SIMAPK5*) with high expression patterns during tomato fruit ripening were screened by tissue expression analysis on <http://bar.utoronto.ca/> (Supplementary Fig. S6).

First, the interaction between *SIWRKY6* and these MAPKs were analyzed by yeast 2-hybrid. As shown in Fig. 4A, only colonies carrying the construct *SIWRKY6*-AD with *SIMAPK4*-BD were grown on selective media, indicating that *SIWRKY6* interacted with

SIMAPK4 in yeast cells. Besides, we further confirmed the interaction between *SIWRKY6* and *SIMAPK4* using pull-down assays. GST-*SIMAPK4* bound to anti-GST agarose beads was mixed with HIS-*SIWRKY6*-purified protein. This result suggested that recombinant HIS-*SIWRKY6* was found to interact with the recombinant GST-*SIMAPK4* in vitro (Fig. 4B). To validate the interaction between *SIWRKY6* and *SIMAPK4* in plant, we performed firefly luciferase complementation (FLC) assays by fusing *SIWRKY6* to the N-terminal end of luciferase (LUC) to form NLuc-*SIWRKY6* and *SIMAPK4* with the C-terminal end of LUC to form CLuc-*SIMAPK4*. We co-infiltrated the NLuc-*SIWRKY6* and CLuc-*SIMAPK4* constructs into *Nicotiana benthamiana* leaves to detect luciferase activity, and the results showed that the luciferase activity of NLuc-*SIWRKY6* and CLuc-*SIMAPK4* was significantly higher than those of the 3 control groups (Fig. 4C). However, when the combination of NLuc-*SIWRKY6* and CLuc-*SIMAPK4* *N. benthamiana* leaves were fumigated with 500 μ M NaHS, the interaction effect was diminished, implying that *SIWRKY6* persulfidation may interfere with the interaction with *SIMAPK4*. The bimolecular fluorescence complementation (BiFC) assay was used to further verify whether this interaction occurs in planta. When co-transformed with *SIWRKY6*-cYFP [yellow fluorescent protein (YFP) carboxy terminal] and *SIMAPK4*-nYFP (YFP amino terminal), fluorescence signals were observed in tobacco cells, suggesting the in vivo interaction between *SIMAPK4* and *SIWRKY6*. In cells transformed with cYFP and nYFP, *SIWRKY6*-cYFP and nYFP, or cYFP and *SIMAPK4*-nYFP, no fluorescence

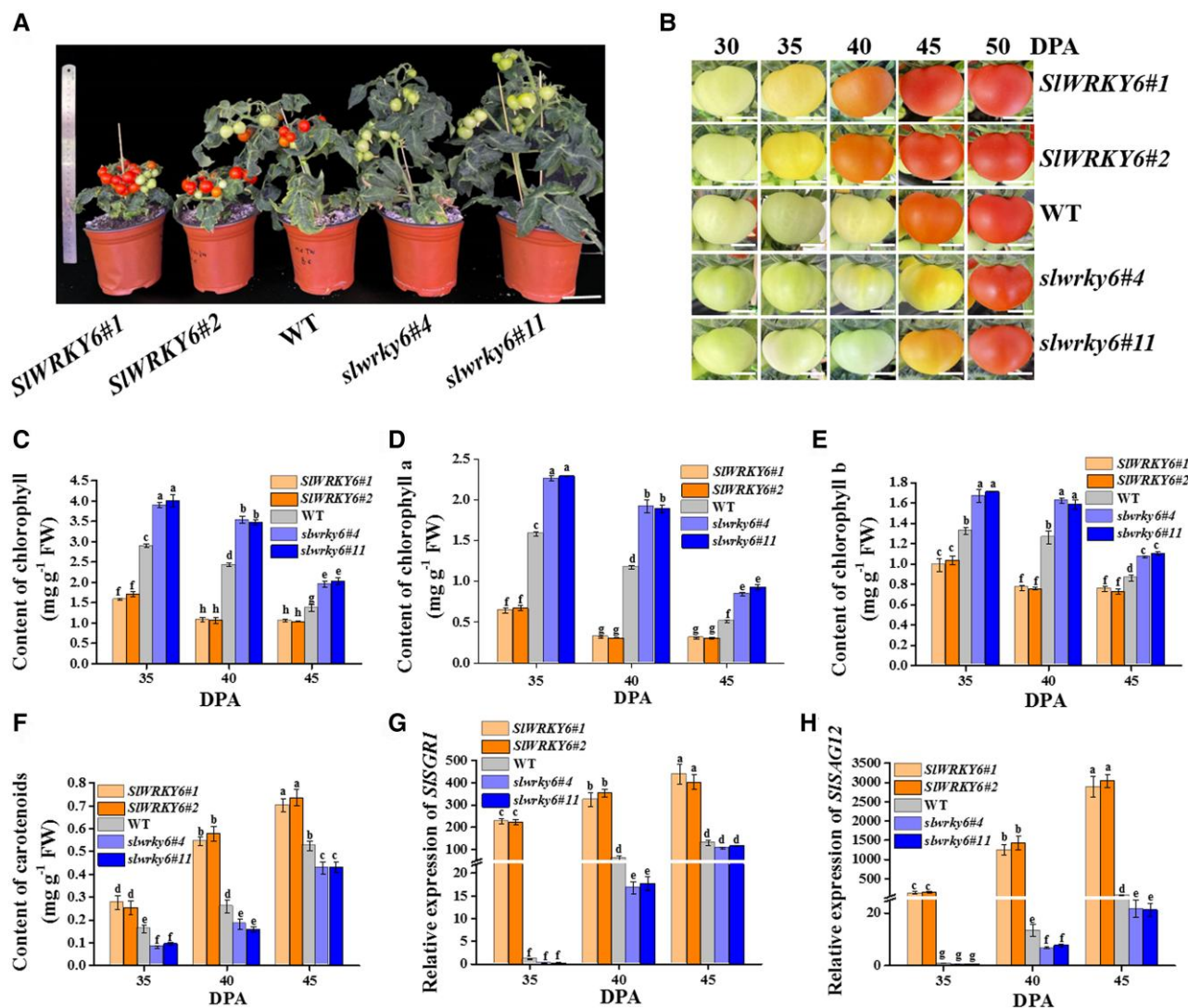


Figure 3. SIWRKY6 affects tomato fruit ripening. **A)** Overall phenotype of *slwrky6* mutants (*slwrky6#4* and *slwrky6#11*), SIWRKY6-OE (SIWRKY6#1 and SIWRKY6#2), and WT tomato plants at 90 d of growth. Image was digitally extracted for comparison. WT, wild type. Scale bars, 6 cm. **B)** Phenotype of tomato fruit ripening in *slwrky6* mutants, SIWRKY6-OE, and WT plants at 30, 35, 40, 45, and 50 DPA. Scale bars, 1 cm. **C) to F)** The contents of total chlorophyll **C)**, Chlorophyll a **D)**, Chlorophyll b **E)**, and carotenoids **F)** in fruits of *slwrky6* mutants, SIWRKY6-OE, and WT plants at 35, 40, and 45 DPA, respectively. FW, fresh weight. **G)** and **H)** The relative expression levels of SISR1 **G)** and SISR12 **H)** were measured in fruits of *slwrky6* mutants, SIWRKY6-OE, and WT plants by RT-qPCR at 35, 40, and 45 DPA. Data are mean $\pm$ SD ($n = 3$). Different letters indicate significant differences between samples ($P < 0.05$, Duncan's test).

signal was detected (Fig. 4D). These results suggested that SIWRKY6 can physically interact with SIMAPK4 in vivo and in vitro, while SIWRKY6 persulfidation damages the interaction.

SIMAPK4 phosphorylates SIWRKY6 at Ser33

To detect whether SIMAPK4 can phosphorylate SIWRKY6, we purified recombinant HIS-tagged SIWRKY6 and GST-tagged SIMAPK4 to perform in vitro kinase assays by Phos-tag mobility shift assay. The protein combination was incubated with lambda protein phosphatase (λ PP) to eliminate the phosphorylation of SIWRKY6. As shown in Fig. 4E, the immunoblot analysis with anti-HIS antibody showed that SIMAPK4 caused an upward shift of the target band of SIWRKY6 while the shift was eliminated by λ PP treatment, indicating that SIMAPK4 were able to phosphorylate SIWRKY6 in vitro.

MAPK family proteins typically phosphorylate serine and threonine residues in the S/T-P motif of their target proteins. By

analyzing the protein sequence of SIWRKY6 by GPS5.0 tool, 4 serine at Positions 33, 229, 248 and 532 (Ser³³, Ser²²⁹, Ser²⁴⁸ and Ser⁵³²) were predicted as potential SIMAPK4 phosphorylation sites. To confirm which sites of serine were phosphorylated sites of SIWRKY6, we mutated each of the 4 Ser sites to the Ala mimic non-phosphorylated form of SIWRKY6 protein. Then in vitro phosphorylation experiments with SIMAPK4 revealed that SIWRKY6 was not phosphorylated by SIMAPK4 when mutating Ser³³ to Ala (S33A) (Fig. 4F), indicating that the Ser33 residue in SIWRKY6 was the main site of phosphorylation by SIMAPK4.

To further investigate which region of SIWRKY6 interacts with SIMAPK4 and the effects of persulfidation and phosphorylation sites on its interaction, different truncated proteins (1 to 293, 294 to 354, and 355 to 550 aa) and mutant proteins of persulfidation and phosphorylation sites (SIWRKY6^{C396A} and SIWRKY6^{S33A}) of SIWRKY6 were subjected to yeast 2-hybrid assays with SIMAPK4. As shown in Supplementary Fig. S7, the 1 to 293 amino acid region of SIWRKY6 could interact with SIMAPK4 and

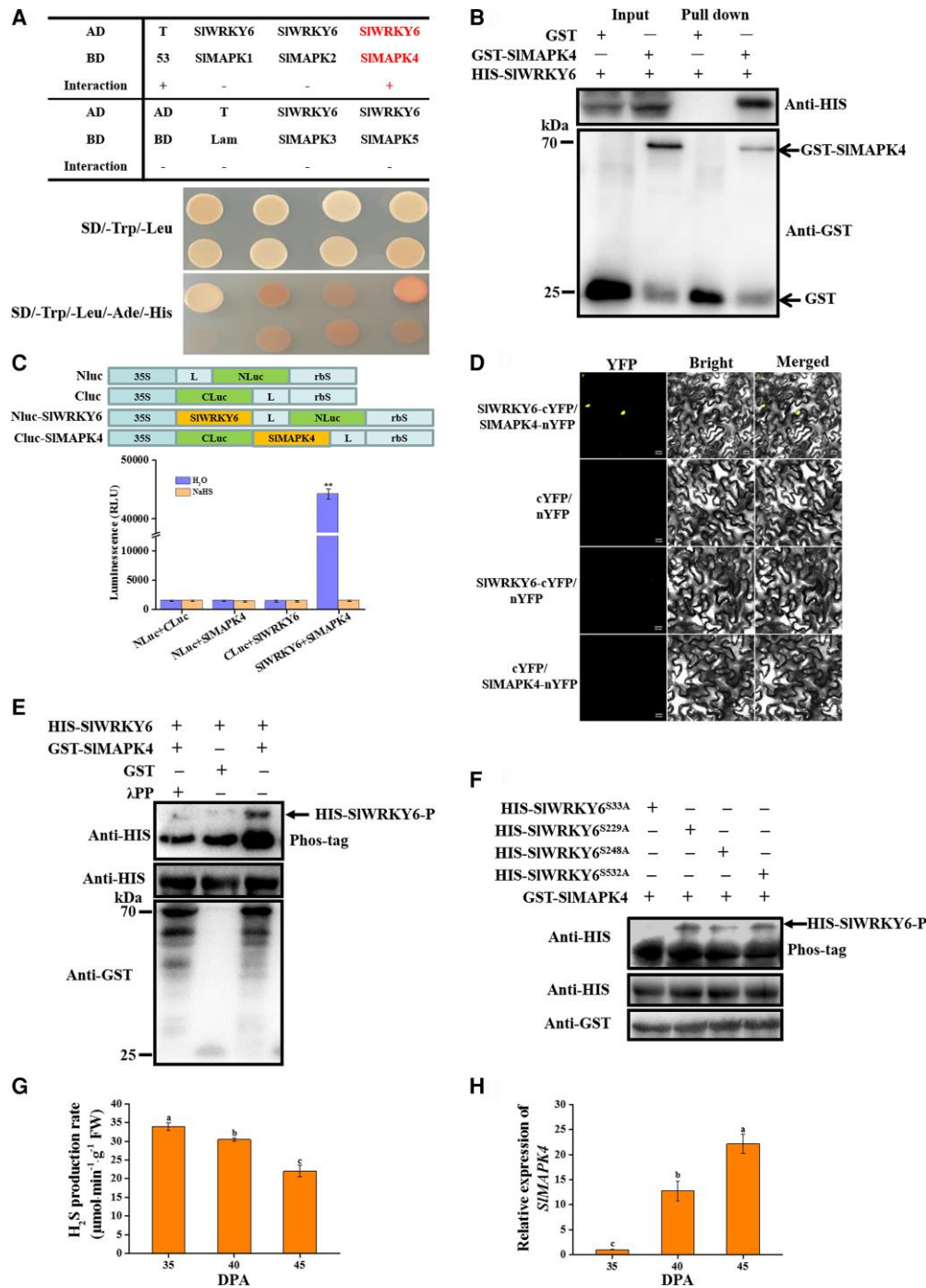


Figure 4. SIWRKY6 physically interacts with SIMAPK4. **A)** The interaction between SIWRKY6 and SIMAPK4 was confirmed by yeast 2-hybrid assay. AD vector expressing AD-SIWRKY6 was co-transformed with BD-SIMAPK4 into yeast cells and cultured in synthetic dropout (SD) medium lacking Leu and Trp (-Leu/-Trp) or SD/-Leu/-Trp/-His/-Ade. Red letters represent positive combinations. AD, pGADT7. BD, pGBKT7. **B)** Pull-down assays of the interaction between SIWRKY6 and SIMAPK4. GST-SIMAPK4 bound to anti-GST agarose beads with HIS-SIWRKY6 purified protein. GST protein was used as a negative control. **C)** The schematic diagram of the NLuc and CLuc vectors. Interaction between SIWRKY6 and SIMAPK4 confirmed by FLC assays in *N. benthamiana* leaves treated with H₂O or 500 μM NaHS. RLU, relative light unit. Data are shown as the means (±SD) of 6 biological replicates. The symbol ** indicates significant difference compared to control ($P < 0.01$, Student's *t*-test). NLuc, pCambia 2300-nLUC vector. CLuc, pCambia 2300-cLUC vector. **D)** SIWRKY6 interacts with SIMAPK4 in vivo by BiFC assays. *N. benthamiana* leaves were cotransformed with SIWRKY6-cYFP and SIMAPK4-nYFP, cYFP and nYFP, SIWRKY6-cYFP and nYFP, or cYFP and SIMAPK4-nYFP. Scale bars, 20 μm. **E)** Phos-tag mobility shift assay showed that SIWRKY6 phosphorylation was mediated by activated SIMAPK4. HIS-SIWRKY6 was detected by immunoblot analysis with anti-HIS antibody and the phosphorylated protein was separated by phos-tag gel. The phosphorylated SIWRKY6 protein migrated slowly on the phos-tag gel (arrows). GST-purified protein and λPPase treatment were used as negative controls. Separate proteins with anti-GST antibody and anti-HIS antibody on a normal SDS-PAGE gel to verify the same loading (bottom). **F)** Analysis of SIWRKY6 mutant forms by phos-tag mobility assay. SIWRKY6 variants harboring alanine substitutions at S33, S229, S248, or S532 (SIWRKY6^{S33A}, SIWRKY6^{S229A}, SIWRKY6^{S248A}, and SIWRKY6^{S532A}). The phosphorylation band was abolished when mutating Ser-33 of SIWRKY6 to Ala. **G)** Endogenous H₂S production in WT fruits at different ripening stages at 35, 40, and 45 DPA. DPA, days post-anthesis; FW, fresh weight. **H)** The relative expression level of SIMAPK4 in WT fruits at 35, 40, and 45 DPA. Error bars are presented as means ± SD (**F**) $n = 5$ and (**G**) $n = 3$). Different letters indicate significant differences between samples ($P < 0.05$, Student's *t*-test). The symbol ** indicates a significant difference compared to other samples ($P < 0.01$, Student's *t*-test).

SIWRKY6 persulfidation site and phosphorylation site mutations did not affect the interaction with SIMAPK4. Correspondingly, the phosphorylation site (S33) of SIWRKY6 located in the region of 1 to 293 aa of SIWRKY6 is also the interaction fragment with SIMAPK4.

To explore the potential relation between H₂S-mediated SIWRKY6 persulfidation and SIMAPK4-mediated phosphorylation, we determined the endogenous H₂S content and SIMAPK4 expression during WT tomato fruit ripening. As shown in Fig. 4, G and H, the endogenous H₂S content decreased and the expression of SIMAPK4 increased as the fruit ripened, implying that SIWRKY6 may be persulfidated in un-ripened fruit and may undergo phosphorylation in ripening fruit.

Effects of WRKY6 persulfidation and phosphorylation on its transcriptional activity to target genes

To examine the effects of SIWRKY6 persulfidation and phosphorylation on its transcriptional activity, we used a dual-luciferase system in *N. benthamiana* leaves to measure the transcriptional activation potential of SIWRKY6 persulfidation and phosphorylation on its downstream target genes SISGR1 and SISAG12. Therefore, we co-infiltrated *N. benthamiana* leaves with effector constructs containing SIWRKY6, SIMAPK4, SIWRKY6^{C396A}, SIWRKY6^{S33A}, and SIWRKY6^{S33D} and different combinations of reporters under the control of the SISGR1 or SISAG12 promoters (Fig. 5A). A dual-LUC reporter assay was performed using *N. benthamiana* leaves co-transformed with the effector SIWRKY6 and the reporter vector constructed using the promoters of SISGR1 (SISGR1_{pro}:LUC) or SISAG12 (SISAG12_{pro}:LUC). The LUC activities of SISGR1_{pro}:LUC and SISAG12_{pro}:LUC were significantly activated by SIWRKY6 and by co-infiltration of SIWRKY6 and SIMAPK4. In the presence of SIMAPK4, co-infiltration of the SIWRKY6 and SISGR1_{pro}:LUC and SISAG12_{pro}:LUC led to significant activation of the fluorescence signal than that of SIWRKY6 alone (Fig. 5, B and C). Relative LUC activity of SISGR1_{pro} or SISAG12_{pro} reporters was significantly higher when SIWRKY6 or SIWRKY6^{C396A} was transiently expressed compared to the pSAK277 control, and transcriptional activity of both promoters continued to increase when co-expressed with SIMAPK4. However, treatment of *N. benthamiana* with NaHS (500 μM) fumigation revealed a significant decrease in LUC activity of SIWRKY6 or SIWRKY6 co-expressed with SIMAPK4, whereas no significant change in LUC activity of SIWRKY6^{C396A} or SIWRKY6^{C396A} co-expressed with SIMAPK4 was observed compared with that of water treatment (Fig. 5, D and E). This result indicated that SIWRKY6 persulfidation modification can inhibit the transcriptional activity of SIWRKY6. Then the effect of SIWRKY6 phosphorylation on its transcriptional activity was evaluated by the phosphonull SIWRKY6^{S33A} and phosphomimetic SIWRKY6^{S33D} vectors. The expression of SIWRKY6^{S33A} resulted in lower transcriptional activity of SISGR1_{pro} or SISAG12_{pro} reporters than SIWRKY6, whereas when a construct expressing phosphomimetic SIWRKY6^{S33D} was co-infiltrated with SIMAPK4 or not in *N. benthamiana* leaves, the LUC activity of the promoters was increased. However, co-expression of SIWRKY6^{S33A} or SIWRKY6^{S33D} and SIMAPK4 did not further increase LUC activity compared with SIWRKY6^{S33A} and SIWRKY6^{S33D} alone (Fig. 5, F and G). The result suggested that phosphorylation of SIWRKY6 by SIMAPK4 was essential for activating the expression of its downstream target genes SISGR1 and SISAG12 during tomato fruit ripening, while SIWRKY6 persulfidation attenuated its transcriptional activity to target genes.

To further clarify the regulatory relationship between SIWRKY6 persulfidation and phosphorylation on its downstream target genes, electrophoretic mobility shift assays (EMSA) were performed using the SIWRKY6 target gene SISGR1 and SISAG12 promoters. The analysis of the SISGR1 and SISAG12 promoters of ~1,000 bp sequences showed that the promoter of SISGR1 had 2 TTGACC binding sequences, while SISAG12 contains 1 TTGACT element (Fig. 6, A and E). As shown in Fig. 6, B and F, it was found that SIWRKY6 protein could only bind to probe 2 of the SISGR1 promoter, though the sequences surrounding the cis-element of the 2 probes were very similar, and could bind to probe of the SISAG12 promoter. The binding ability of SIWRKY6 protein with both promoters was diminished when unlabeled promoter fragments were added as competitors. When SIWRKY6 was modified by persulfidation mediated by NaHS treatment, the binding ability to SISGR1 and SISAG12 promoters became weaker with increasing NaHS concentration as evidenced by the weaker intensity of the retarded bands (Fig. 6, C and F). When SIWRKY6 was phosphorylated by SIMAPK4, the binding affinity of phosphorylated SIWRKY6 for the target gene promoters was clearly enhanced due to the stronger intensity of the bands (Fig. 6, D and G). The above results indicated that SIWRKY6 persulfidation inhibited its binding ability to the W-box of SISGR1 and SISAG12 promoters, while its phosphorylation by SIMAPK4 enhances its binding activity to target genes.

To explore the roles of SISGR1 and SISAG12 in regulating tomato fruit ripening, we silenced the expression of SISGR1 and SISAG12 in WT by virus-induced gene silencing (VIGS). SISGR1 and SISAG12 silencing in WT caused delayed fruit ripening compared to pTRV control. In addition, pTRV-SISGR1 and pTRV-SISAG12 in WT displayed higher chlorophyll (including Chlorophyll a and Chlorophyll b) content and lower carotenoid content than control. Besides, the expression level of SISGR1 and SISAG12 was significantly lower in pTRV-SISGR1 and pTRV-SISAG12 than in the counterpart controls (Supplementary Fig. S8). Therefore, the result suggested that SISGR1 and SISAG12 are positive regulators for fruit ripening in tomato.

Effect of SIWRKY6 persulfidation and phosphorylation in regulating tomato fruit ripening

To investigate the effects of SIWRKY6 modification by persulfidation and phosphorylation in regulating fruit ripening of tomato, we transiently expressed SIWRKY6, SIWRKY6^{C396A} which could not be persulfidated, phosphonull SIWRKY6^{S33A}, and phosphomimetic SIWRKY6^{S33D} by pSAK277 vector in *slwrky6* mutant fruits. The tomatoes transfected with SIWRKY6 and SIWRKY6^{C396A} turned red earlier compared to pSAK277 control (Fig. 7A). However, tomato fruits ripened slower after H₂S fumigation by NaHS than H₂O treatment, and the ripening process was a little accelerated in SIWRKY6^{C396A} compared with SIWRKY6 under H₂S fumigation. SIWRKY6^{S33A} became red on the 7th day after injection, and SIWRKY6^{S33D} fruit color was a little redder compared with SIWRKY6 (Fig. 8A). Next, we determined the total chlorophyll, Chlorophyll a, Chlorophyll b, and carotenoid contents of the fruits, and generally, the parameters were in agreement with the phenotype caused by transient expression (Figs. 7, B to E and 8, B to E). The expression levels of SIWRKY6 and SIWRKY6^{C396A} were significantly higher in transient expression groups than in the counterpart controls (Fig. 7F), and the expressions of SIWRKY6, SIWRKY6^{S33A}, and SIWRKY6^{S33D} significantly higher than those of control (Fig. 8F). As shown in Fig. 7, G and

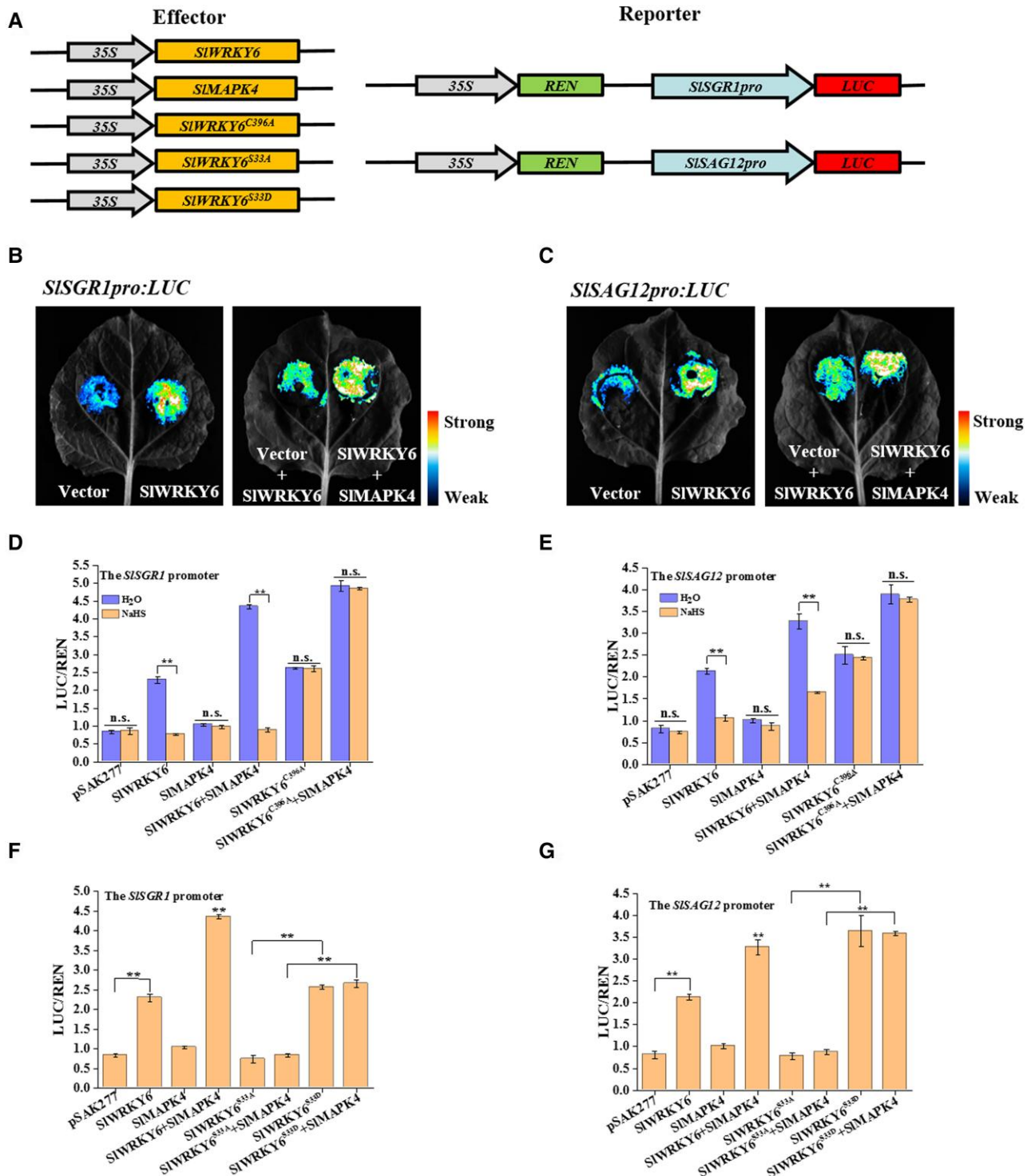


Figure 5. Effect of SIWRKY6 persulfidation and phosphorylation on the transcriptional activity of SIWRKY6. **A)** Schematic diagram of the effector vectors expressing SIWRKY6, SIWRKY6^{C396A}, SIWRKY6^{S33A}, and SIMAPK4 and the reporter constructs containing the SiSGR1 and SiSAG12 promoters for the dual-luciferase reporter assay. REN, renilla luciferase. LUC, firefly luciferase. **B)** and **C)** Activity assay of SIWRKY6 in regulating SiSGR1 or SiSAG12. Luciferase reporter assay showed the binding of SIWRKY6 to the SiSGR1 or SiSAG12 promoter in vivo. The infected *N. benthamiana* was visualized by a living fluorescence imager with D-luciferin (sodium salt). The color scale bar indicates the range of luminescence intensity. **D)** and **E)** Dual-luciferase assay (LUC) showed the transcriptional activity of SIWRKY6, SIWRKY6^{C396A} and SIMAPK4 on the transcription of downstream target gene SiSGR1 **D)** and SiSAG12 **E)**. LUC/REN ratios indicated the transactivation effect to SiSGR1 and SiSAG12 promoters. *N. benthamiana* leaves which expressing the vectors were treated with H₂O and 500 μ M NaHS, respectively, to elucidate the effect of SIWRKY6 persulfidation or SIWRKY6^{C396A} which could not be persulfidated. **F)** and **G)** Effect of phosphorylation of SIWRKY6 on the transcription of SiSGR1 **F)** and SiSAG12 **G)** promoters by LUC experiment in *N. benthamiana* leaves, respectively. The effector vectors included SIWRKY6^{S33A}, mutation of serine (S) to alanine (A) at Position 33 which prevents phosphorylation of SIWRKY6. SIWRKY6^{S33D}, mutation of serine (S) to aspartate (D) mimics phosphorylation and SIMAPK4 in single or in combination. Data are shown as the means ($\pm$ SD) of 6 biological replicates. Stars are used to indicate significant differences (** $P < 0.01$, Student's t-test) and n.s. stands for no significant difference.

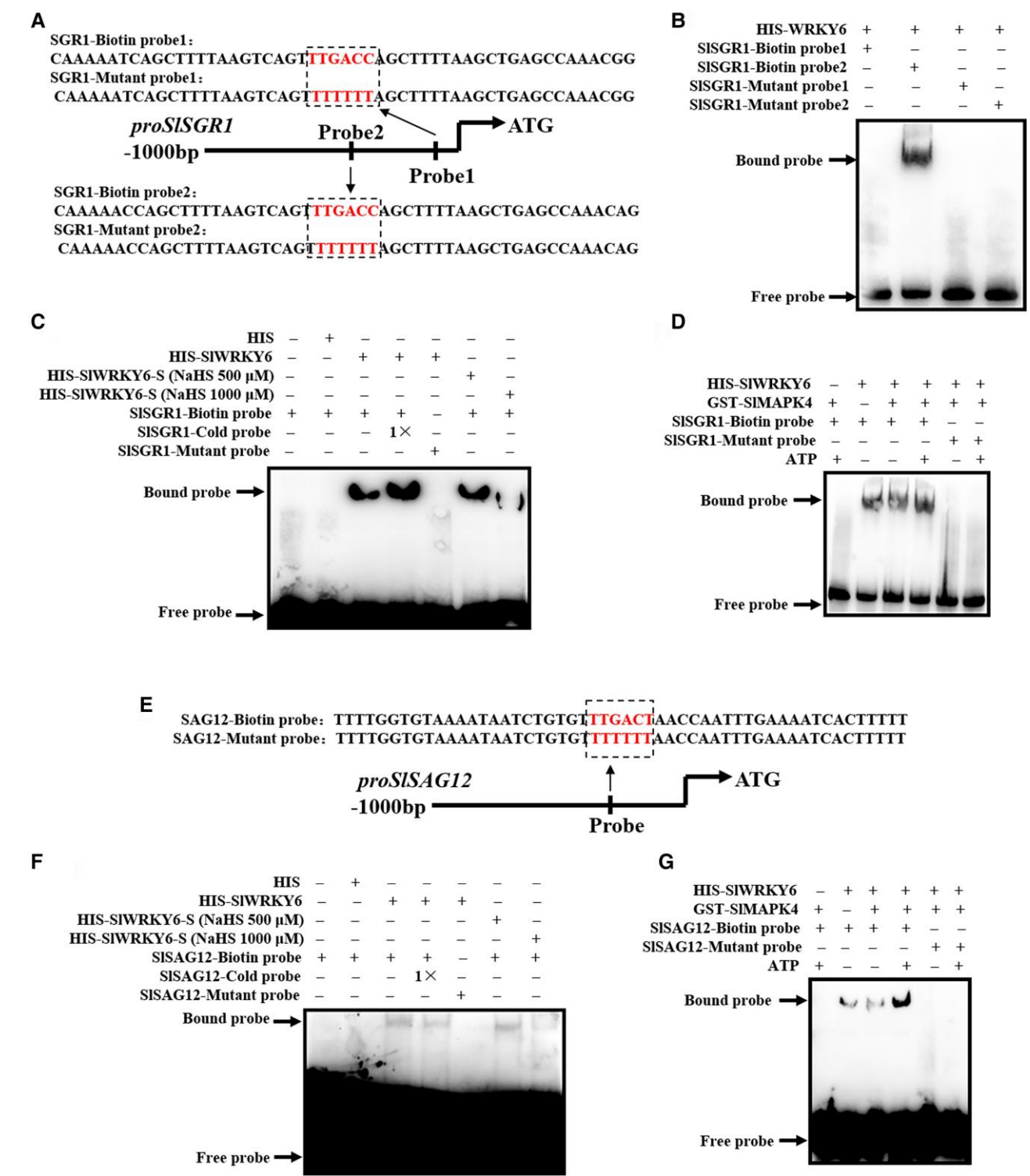


Figure 6. DNA-binding activity of SIWRKY6 to the SISGR1 and SISAG12 promoters was assessed by EMSA. **A)** and **E)** Analysis of WRKY binding sites of SISGR1 and SISAG12 promoters in -1,000 bp. **B)** Identification of the DNA-binding element of SIWRKY6 in SISGR1 promoter. **C)** and **F)** The SIWRKY6 protein was treated with 500 or 1,000 μM NaHS for 1 h and the DNA-binding ability to SISGR1 **C)** and SISAG12 **F)** probes were evaluated by EMSA. **D)** and **G)** EMSA showing the binding ability of SIWRKY6 phosphorylation to SISGR1 **D)** and SISAG12 **G)** promoters. The proteins include purified HIS-SIWRKY6 or HIS-SIWRKY6 in combination with GST-SIMAPK4. The cold probes are unlabeled fragments used as competitive to probes (1x). The mutant probes are biotin-labeled probes with mutations from TTGACC/T to AAAAA (red letters), -, absence, +, presence.

H, the transcript levels of SISGR1 and SISAG12 were upregulated in SIWRKY6 and SIWRKY6^{C396A} fruits, but H₂S fumigation attenuated the upregulation of these genes. Besides, the expression levels of SISGR1 and SISAG12 were upregulated in SIWRKY6 and

SIWRKY6^{S33D}, while SIWRKY6^{S33A} attenuated the upregulation of SISGR1 and SISAG12 expression (Fig. 8, G and H). In all, the data suggested that SIWRKY6 persulfidation inhibited its transcriptional activity and affected the binding of SIWRKY6 to

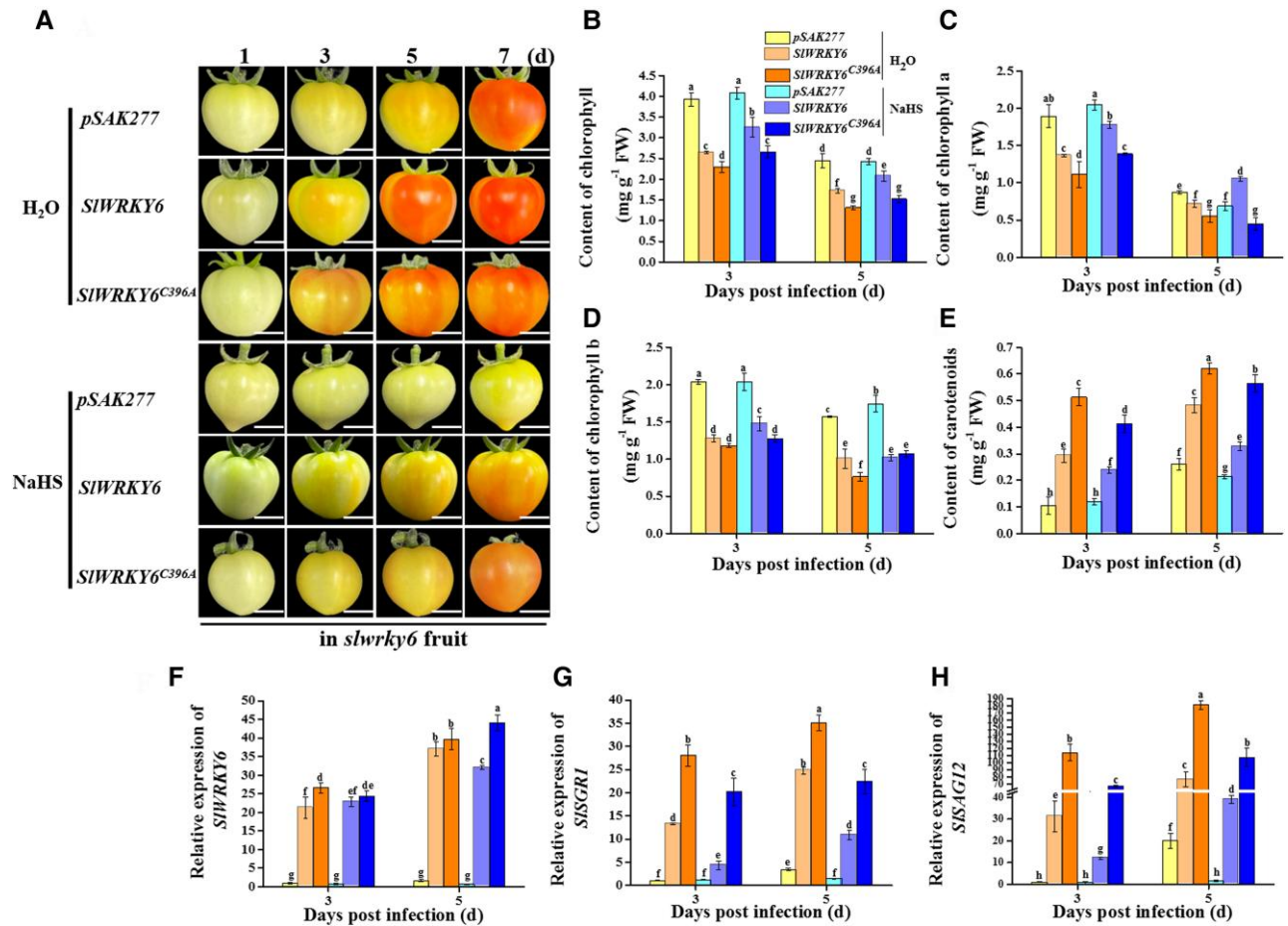


Figure 7. Effect of SIWRKY6^{C396A} mutation on tomato fruit ripening detected by transient expression under NaHS treatment in *slwrky6* fruits. **A)** The *slwrky6* mutant fruits were injected with pSAK277 (control), SIWRKY6 or SIWRKY6^{C396A}-pSAK277 at the green ripening stage (25 DPA) and treated with H₂O or 500 μ M NaHS to observe fruit phenotypic changes after storage at 25°C for 1, 3, 5, and 7 d. Images were digitally extracted for comparison. Scale bars, 1 cm. **B) to E)** The contents of total chlorophyll **B)**, Chlorophyll a **C)**, Chlorophyll b **D)**, and carotenoids **E)** in tomato fruits in **A)** at 3 and 5 d of storage, respectively. FW, fresh weight. **F) to H)** The expression levels of SIWRKY6 **F)**, SISGR1 **G)**, and SISAG12 **H)** in fruits in **A)** by RT-qPCR. Error bars are presented as means $\pm$ SD ($n = 3$). Different letters indicate significant differences between samples ($P < 0.05$, Duncan's test).

SISGR1 and SISAG12 promoters, thereby inhibiting tomato fruit ripening, whereas SIWRKY6 phosphorylation activated downstream target genes and thus promoted tomato ripening.

SlMAPK4 is required for the function of SIWRKY6 in fruit ripening

To further explore the genetic relationship between SlMAPK4 and SIWRKY6, we silenced SlMAPK4 in WT and SIWRKY6#1 fruits by VIGS. SlMAPK4 silencing in WT caused delayed fruit ripening, and SlMAPK4 silencing in SIWRKY6 overexpression fruit also retarded fruit ripening compared to pTRV control (Fig. 9A). In addition, pTRV-SlMAPK4 in WT and pTRV-SlMAPK4 in SIWRKY6 fruit displayed higher chlorophyll (including Chlorophyll a and Chlorophyll b) content and lower carotenoid content than WT: pTRV and SIWRKY6#1: pTRV (Fig. 9, B to E). Finally, the relative expressions of SlMAPK4, SIWRKY6, SISGR1, and SISAG12 were examined by RT-qPCR. The expression level of SlMAPK4 was significantly lower in pTRV-SlMAPK4 and SIWRKY6#1: pTRV-SlMAPK4 than in the counterpart controls (Fig. 9F). The expression level of SIWRKY6 is not much affected by SlMAPK4 silencing (Fig. 9G). However, the expression levels of SISGR1 and SISAG12 were much lower in pTRV-SlMAPK4 and SIWRKY6#1: pTRV-SlMAPK4 compared with WT: pTRV and SIWRKY6#1: pTRV (Fig. 9, H and I). Collectively, these results indicated that SlMAPK4 played

a positive regulatory role in fruit ripening by phosphorylating SIWRKY6. Finally, we provide a model based on the persulfidation and phosphorylation of SIWRKY6 in balancing tomato fruit ripening (Fig. 10).

Discussion

Tomato fruit ripening is a programmed physiological process involving a variety of physiological changes including chlorophyll degradation and carotenoid synthesis, etc. During fruit ripening or leaf senescence, SGR1 plays an important role in activating chlorophyll degradation and in tomato SGR1 deletion line shows higher chlorophyll and carotenoid levels (Kim et al. 2022). SAG12 encodes a cysteine protease, participating in macromolecule degradation and is upregulated during plant senescence (Doan et al. 2022). TFs, including WRKY, ERFs, and NACs, have been shown to bind directly to the promoters of chlorophyll degradation and senescence-related genes, such as PPH, SGR1, NYC, and SAG12, and can regulate their transcription, which is consistent with their involvement in the regulation of fruit ripening and leaf senescence (Li et al. 2021; Zhang et al. 2021; Wang et al. 2022). Interestingly, Yuan et al. (2022) showed that SIWRKY35 promotes fruit ripening by positively regulating carotenoid biosynthesis through activation of the 2-c-methyl-D-erythritol 4-phosphate pathway in tomato fruit. LcNAC002 was shown to promote

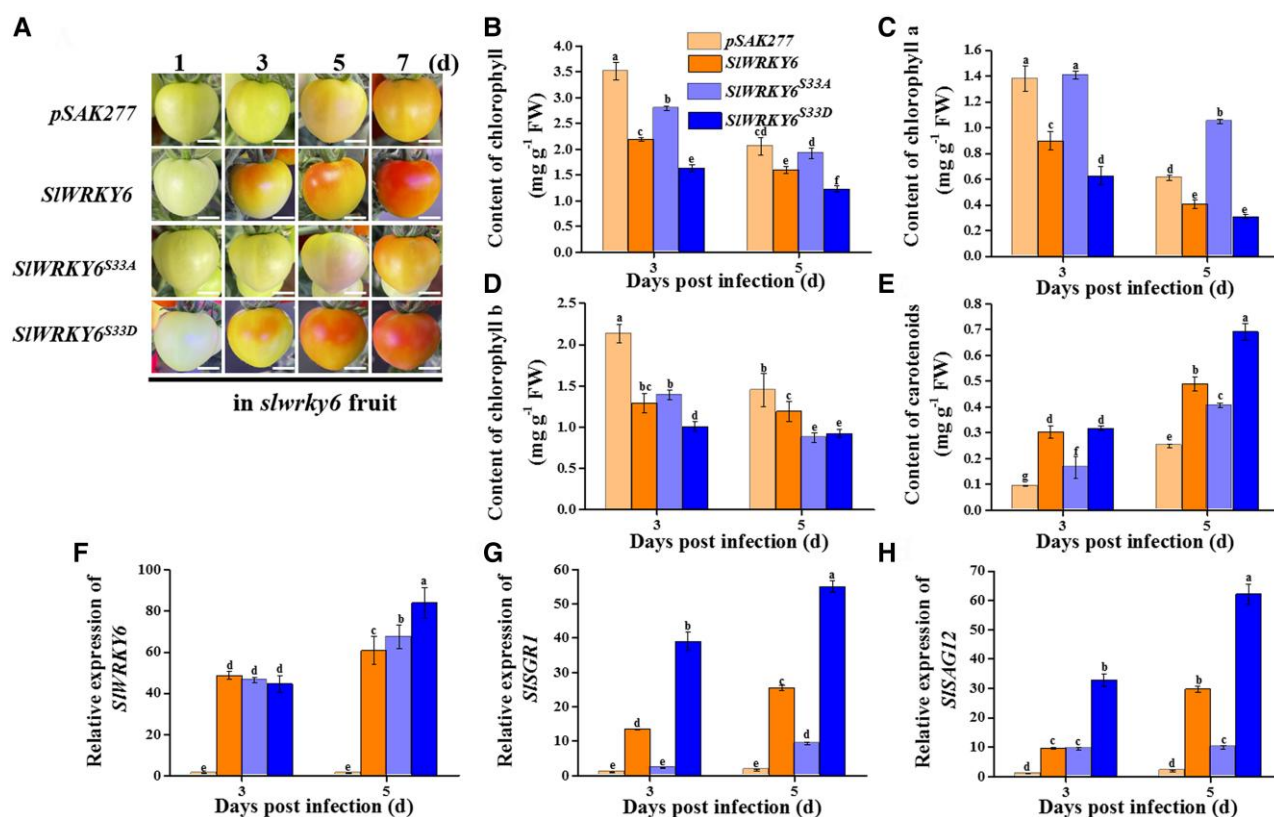


Figure 8. Effect of SIWRKY6 serine (S^{33}) to alanine (A) mutation which could not be phosphorylated and serine (S^{33}) to aspartate (D) mimics phosphorylation mutation on tomato fruit ripening as detected by transient expression in *slwrky6* fruits. **A)** The *slwrky6* mutant fruits were injected with pSAK277 (control) or SIWRKY6^{S33A}-pSAK277 or SIWRKY6^{S33D}-pSAK277 at green ripening stage and fruit phenotypic changes were observed after storage at 25°C for 1, 3, 5, and 7 d. Scale bars, 1 cm. **B) to E)** The contents of total chlorophyll **B)**, Chlorophyll a **C)**, Chlorophyll b **D)**, and carotenoids **E)** in tomato fruits, respectively. FW, fresh weight. **F) to H)** The expression levels of SIWRKY6 **F)**, SISGR1 **G)**, and SISAG12 **H)** in fruit by RT-qPCR. Error bars are presented as means $\pm$ SD ($n = 3$). Different letters indicate significant differences between samples ($P < 0.05$, Duncan's test).

chlorophyll degradation by directly targeting the LcSGR promoter (Zou et al. 2023). In this study, through dual-luciferase transient expression assays and electrophoresis mobility shift assays, we confirm the function of WRKY TF SIWRKY6 in promoting tomato fruit ripening, by directly promoting the transcription of ripening and senescence-related genes SISGR1 and SISAG12.

Post-translational modifications of proteins may alter the structure and/or conformation of proteins, thereby activating or inhibiting enzyme activity, changing their interaction ability with other molecules (proteins, DNA and RNA), affecting their subcellular localization or changing their stability (Kwon et al. 2006). Increasing evidence have highlighted that protein modifications such as phosphorylation, ubiquitination, persulfidation, and glycosylation regulated fruit ripening (Sun et al. 2023; Wei et al. 2023; Xu et al. 2023). H_2S can post-translationally modify -SH in protein cysteine residues to form -SSH group through persulfidation and is considered as an important way for regulating protein function. For instance, H_2S induces S-sulfhydryl modification of the Cys131 and Cys137 sites of the Arabidopsis SnRK2.6 protein, which promotes protein kinase and its substrate protein ABF2 which in turn promotes ABA-induced stomatal closure (Chen et al. 2020). In the present work, we find that SIWRKY6 undergoes differential protein persulfidation in *SILCD1* overexpression leaves, suggesting that SIWRKY6 may be a target protein for H_2S -mediated persulfidation. This study reveals that SIWRKY6 at Cys396 can be modified by persulfidation in vitro which significantly inhibits fruit ripening, but SIWRKY6^{C396A} which cannot be persulfidated eliminates this inhibitory effect. Further validation

shows that SIWRKY6 persulfidation delays tomato fruit ripening by attenuating the transcriptional activity to the target genes SISAG12 and SISGR1. Robatzek and Somssich (2002) show that AtWRKY6 could directly activate SIRC to accelerate leaf senescence in Arabidopsis. Similarly, we also find that SIWRKY6 over-expression accelerates both leaf senescence and fruit ripening. To explore the mechanism of SIWRKY6 persulfidation in regulating fruit ripening, dual-luciferase system in *N. benthamiana* leaves is applied to measure the transcriptional activation potential of SIWRKY6 persulfidation. H_2S fumigation of *N. benthamiana* leaves reveals a significant decrease in the LUC activity of SIWRKY6 or SIWRKY6 co-expressed with SIMAPK4, whereas no significant change in LUC activity of SIWRKY6^{C396A} or SIWRKY6^{C396A} co-expressed with SIMAPK4 is observed compared with that of water treatment (Fig 5, D and E), suggesting that SIWRKY6 persulfidation modification can inhibit the transcriptional activity of SIWRKY6. Further, by EMSA experiment, SIWRKY6 treated with H_2S which leads to its persulfidation weakens the binding ability to SISGR1 and SISAG12 promoters as evidenced by the weaker intensity of the retarded bands. SIWRKY6 modification by persulfidation in regulating tomato fruit ripening is also confirmed by transient expression of SIWRKY6, SIWRKY6^{C396A} which could not be persulfidated in *slwrky6* mutant fruits. SIWRKY6 persulfidation in tomato fruit in vivo reveals that mutation of cysteine at Position 396 of SIWRKY6 results in accelerated fruit ripening, increased carotenoid content, and increased expression of target genes SISGR1 and SISAG12. Collectively, our study showed that SIWRKY6 persulfidation represses its transcriptional regulatory

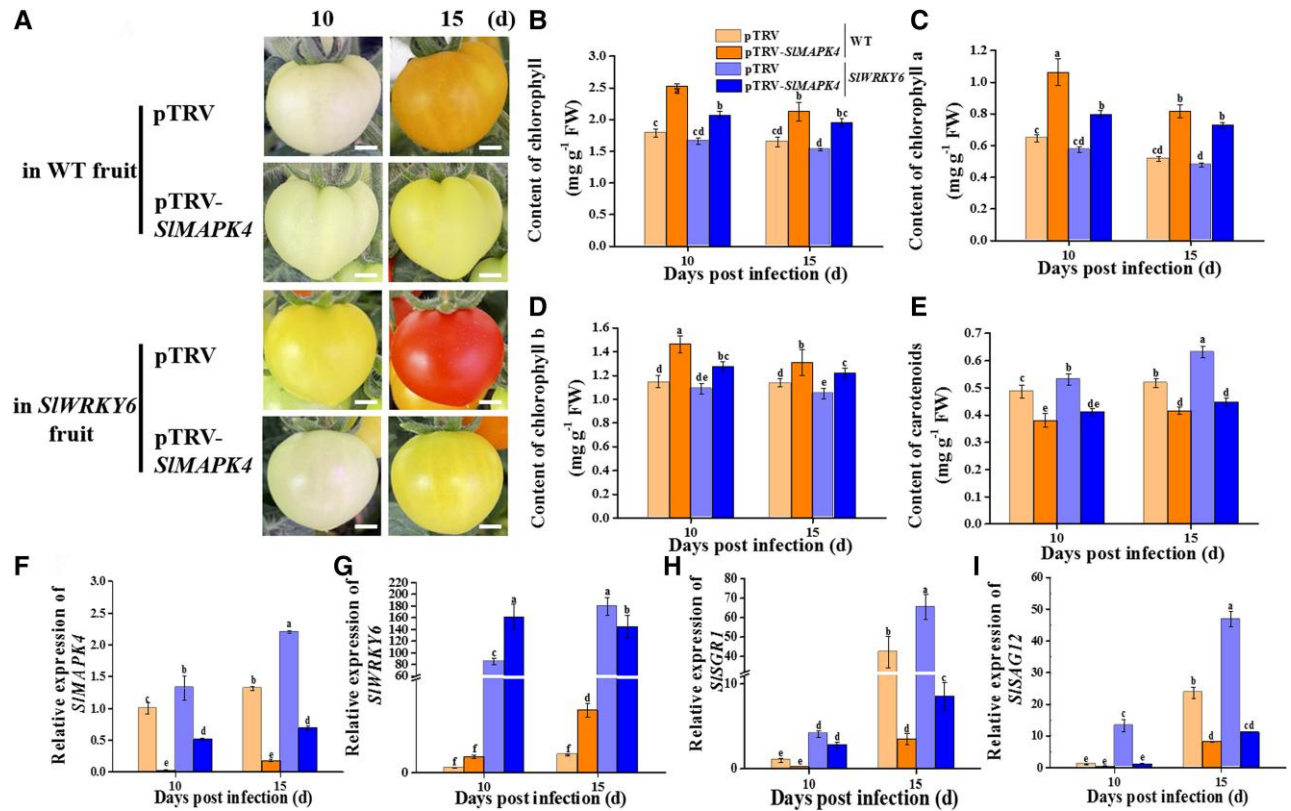


Figure 9. Effect of SIMAPK4 on the function of SIWRKY6 in fruit ripening by VIGS. **A)** Phenotypes of fruits infected with pTRV or pTRV-SIMAPK4 in WT and SIWRKY6#1 plants, at 10 and 15 d post injection. WT, wild type. Scale bars, 0.5 cm. **B) to E)** The content of total chlorophyll **B)**, Chlorophyll a **C)**, Chlorophyll b **D)**, and carotenoids **E)** at 10 and 15 d post-injection, respectively. FW, fresh weight. **F) to I)** The expression levels of SIMAPK4 **F)**, SIWRKY6 **G)**, SISGR1 **H)**, and SISAG12 **I)** in fruits in **A)** by RT-qPCR. Data are presented as means $\pm$ SD ($n = 3$). Different letters indicate significant differences between samples ($P < 0.05$, Duncan's test).

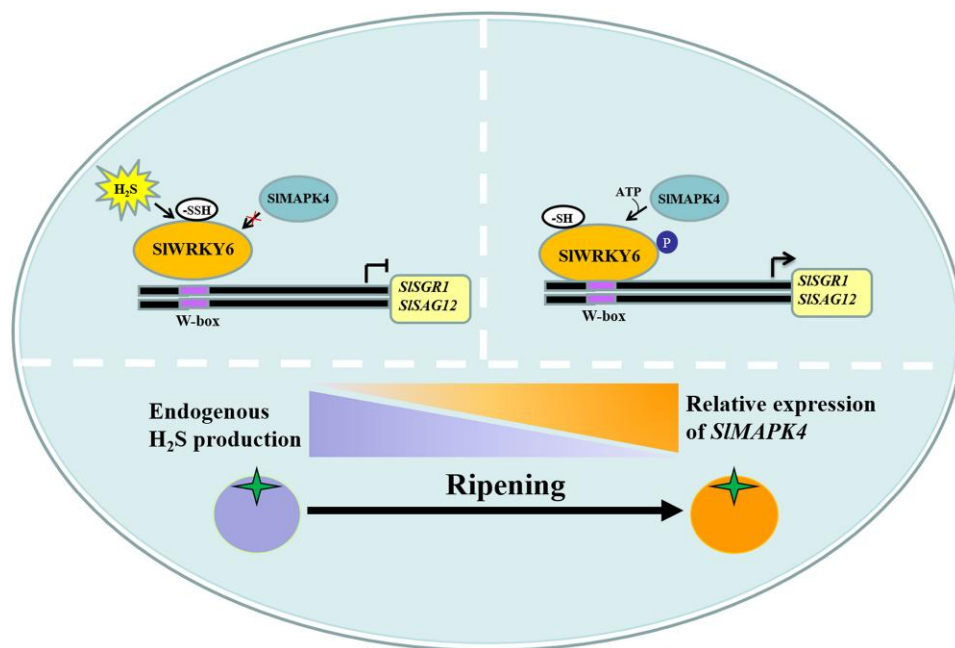


Figure 10. The model showing the molecular mechanism of differential roles of persulfidation and phosphorylation of SIWRKY6 in regulating tomato fruit ripening. In the early stage of fruit ripening, SIWRKY6 was modified by persulfidation due to high levels of endogenous H₂S in un-ripened fruit, which may inhibit its phosphorylation by SIMAPK4, which in turn inhibited the transcription of SISGR1 and SISAG12. As the fruit ripened, H₂S production decreased and SIMAPK4 expression increased, and thus phosphorylated SIWRKY6 increased the transcriptional activity of SISGR1 and SISAG12, leading to fruit ripening. -SSH, persulfidated state; -SH, sulfhydryl group; P, phosphate group.

activity to target genes. Similarly, Yao et al. (2023) reported that PyMYB10 persulfidation inhibits anthocyanin synthesis in red-skinned pears.

Post-translational modifications are essential for protein function. Accumulating evidence show that many WRKY family proteins can undergo phosphorylation as substrates of MAPKs in Arabidopsis (Guan et al. 2014), rice (Wang et al. 2023), and cotton (Wang et al. 2021), thereby regulating plant growth and development. Previous studies revealed that AtMPK3 and AtMPK6 were involved in leaf senescence and floral organ abscission in Arabidopsis (Yue et al. 2012; Guan et al. 2014). Besides, studies have demonstrated that the MAPK cascade is involved in the ethylene signaling pathway. For instance, Arabidopsis MAPK6 can increase ethylene biosynthesis by phosphorylating ACS2/ACS6 to enhance its protein stability and activity (Liu and Zhang 2004). Ethylene was found to induce enhancement of MdMAPK3 phosphorylation of MdNAC72, thereby promoting apple fruit softening (Wei et al. 2023). MAPK3 and MAPK6 can regulate the expression of ACS2 and ACS6 genes by phosphorylating the TF WRKY33 (WRKY DNA-binding protein 33) in Arabidopsis (Li et al. 2012). Although phosphorylation is known to be essential for the ethylene synthesis pathway in fruit ripening by promoting the stability of TFs, there are fewer reports on the effects of TF phosphorylation on the chlorophyll degradation pathway. In the present work, SIMAPK4 is found to interact with SIWRKY6 and to phosphorylate SIWRKY6 at Ser33 by in vitro phosphorylation experiments. More interestingly, SIWRKY6 phosphorylation can directly activate the expression of its downstream target genes SISGR1 and SISAG12 (Fig. 5, F and G). This result is also verified by EMSA as when SIWRKY6 is phosphorylated by SIMAPK4, the binding affinity of phosphorylated SIWRKY6 for the target gene promoters is clearly enhanced due to the stronger intensity of the bands (Fig. 6, D and G). Also, how SIWRKY6 phosphorylation regulates tomato fruit ripening is further investigated by transient expression of SIWRKY6, phosphonull SIWRKY6^{S33A} and phosphomimetic SIWRKY6^{S33D} by pSAK277 vector in *slwrky6* mutant fruit. The results indicates that SIWRKY6^{S33D} accelerates tomato ripening, while the opposite is found for SIWRKY6^{S33A}. Therefore, SIMAPK4 acts as a molecular bridge linking chlorophyll degradation and senescence gene expression to fruit color transition during tomato fruit ripening. Moreover, SIMAPK4 is important for the phosphorylation of SIWRKY6, and phosphorylation of SIWRKY6 by SIMAPK4 activates the expression of SIWRKY6 downstream target genes, thereby promoting the ripening of SIWRKY6 tomato fruits.

The relations between SIWRKY6 persulfidation and phosphorylation are also explored. When the combination of NLuc-SIWRKY6 and CLuc-SIMAPK4 *N. benthamiana* leaves are fumigated with H₂S, the interaction effect is diminished, implying that SIWRKY6 persulfidation may interfere the interaction with SIMAPK4. Besides, it is found that as tomato fruit ripens, endogenous H₂S production decreases, while SIMAPK4 expression increases, suggesting the changes may alter the post-translational modification of SIWRKY6. In the EMSA experiment, SIWRKY6 persulfidation by H₂S in combination with SIMAPK6 phosphorylation weakens its binding ability to the W-box of SISGR1 and SISAG12 promoters compared with control SIWRKY6, also implying that SIWRKY6 persulfidation may interfere with its phosphorylation. Therefore, our data provides insights into the mechanism of persulfidation and phosphorylation of SIWRKY6 in regulating tomato fruit ripening, which may improve the understanding of the molecular basis by which H₂S inhibits fruit ripening.

Materials and methods

Plant materials and growth conditions

The WT tomato (*S. lycopersicum* "Micro-Tom") used in this study and transgenic lines constructed in this background (2 CRISPR-Cas, 9 mutant lines *slwrky6#4* and *slwrky6#11*, and 2 pBI121-FLAG lines SIWRKY6#1 and SIWRKY6#2) were grown in a nutrient soil: vermiculite (3:1, v/v) in pots of 10 cm diameter with 16-h light/8-h dark cycles at 25°C.

H₂S fumigation of tomato fruits

The WT tomato fruits at the green ripening stage were fumigated in a 10 L closed container with 100 mL 500 μM NaHS as H₂S donor or water at 25°C. Random 10 tomato fruits with similar sizes and shapes were selected and each group contained 3 replicates. The NaHS solution or water was changed every 24 h and photographs were taken daily for phenotypic observation.

Plasmid construction and plant transformation

To construct SIWRKY6 loss-of-function mutant plants, 2 gRNA sequences of SIWRKY6 were designed by the CRISPRdirect website (<http://crispr.dbcls.jp/>) (Supplementary Table S1). The synthesized gRNAs were annealed and integrated with AtU3d and AtU3b promoter which were further inserted into the pYLCRISPR/Cas9-DN binary vector. To overexpress SIWRKY6, the coding sequence (CDS) of SIWRKY6 without the stop codon were amplified from the cDNA of the WT tomato and inserted into the pBI121-FLAG vector at the sites of XbaI and XhoI. The recombinant plasmids were transformed into *Agrobacterium tumefaciens* EHA105 and the correct strains were transformed into tomato cv. Micro-Tom by *Agrobacterium*-mediated cotyledon tissue culture. All the mutant plants were genotyped by Sanger sequencing of the PCR products of genomic DNA flanking the target site (Supplementary Table S2). SIWRKY6 overexpression plants were confirmed by RT-qPCR and Western blotting with anti-FLAG antibody. Two independent homozygous lines of *slwrky6* mutants and SIWRKY6 overexpression of the T2 generation plants were used for further study.

Identification of persulfidated proteins by LC-MS/MS

The persulfidation proteins were identified according to the method of Aroca et al. (2017). Tomato leaves of WT and *SILCD1* were lysed in the lysis solution of 4% (w/v) SDS and 100 mM Tris/HCl at pH 7.6 to extract proteins, followed by protein quantification by bicinchoninic acid method. An appropriate amount of protein from each sample was trypsinized by filter-aided proteome preparation, and the peptides were desalted by C18 Cartridge, lyophilized in 40 μL of 0.1% (v/v) formic acid solution, and then quantified by peptide quantification (OD280). The peptides were analyzed by liquid chromatography and MS. MS and MS/MS data obtained for individual samples were processed using Analyst TF 1.5.1 Software (AB SCIEX).

Biotin-switch detection of protein persulfidation

The persulfidation of SIWRKY6 protein was detected by modified biotin-switch according to the method of Sun et al. (2023). The coding sequence of SIWRKY6 or SIWRKY6 with mutation from Cys to Ala at Position 33 amplified by over-lap PCR was cloned onto pCold vector to generate HIS-SIWRKY6. Recombinant His-SIWRKY6 were expressed in *E. coli* Rosetta (DE3) by induction with 0.5 mM isopropyl β-D-1-thiogalactopyranoside

(IPTG) overnight at 16°C for 16 h. HIS-SIWRKY6 protein was purified by Ni Sepharose 6 Fast Flow (GE, 17531806). Then 200 μ L of purified HIS-SIWRKY6 protein was treated with 0 μ M, 500 μ M, 1,000 μ M H_2S , or 1 mM DTT at 4°C for 1 h. The protein was labeled the biotin-switch method. Then the protein samples were separated on 10% (v/v) SDS-PAGE gel, and after electrophoresis, the proteins were transferred onto 0.45 μ M polyvinylidene difluoride membranes by Western blotting with anti-biotin rabbit polyclonal antibody as the primary antibody. The primers used are listed in [Supplementary Tables S3 and S4](#).

Persulfidation site analysis of SIWRKY6 by LC-MS/MS

The persulfidation sites of SIWRKY6 protein were assayed by LC-MS/MS. The purified SIWRKY6 protein of 200 μ L was treated with 1,000 μ M NaHS at 4°C for 1 h. Then, the sample was analyzed by LC-MS/MS (FusionLumos, ThermoFisher, USA) as described by [Chen et al. \(2021\)](#).

Transcriptome analysis

Three biological replicates of mature leaf samples of WT and SIWRKY6 overexpression plants grown for 60 d were collected for RNA-seq analysis. RNA-seq was performed at OE Biotech (China) as described by [Wu et al. \(2020\)](#). The clean reads were aligned to the Tomato Genome version SL4.0. The DEGs between WT and SIWRKY6 overexpression leaves were analyzed under fold change ≥ 2 and false discovery rate < 0.05 . KEGG pathway of DEGs was analyzed using (<https://www.genome.jp/kegg/>).

Measurement of chlorophyll and carotenoid in tomato leaves and fruits

The contents of chlorophyll and carotenoid were determined by the method of [Yao et al. \(2020\)](#). Fresh tomato sample of 2.0 ± 0.01 g was ground in liquid nitrogen and extracted with 95% (v/v) ethanol. The absorbance was measured at 665, 649, and 470 nm. Three biological replicates were performed for each sample, and each biological replicate includes 3 technical replicates. The results were expressed as mg/g FW (fresh weight).

RNA extraction and reverse transcription quantitative PCR (RT-qPCR) analysis

Total RNA was extracted using FastPure Universal Plant Total RNA Isolation Kit (Vazyme, China). Then, cDNA synthesis was performed according to Evo M-MLV One-step RT-PCR Kit (AG, China). RT-qPCR was performed by the manufacturer's instructions of Light Cycler 96 SYBR GREEN I Master (Roche, USA). The relative expression level of genes was analyzed using the $2^{-\Delta\Delta Ct}$ analysis method by normalizing to the reference gene *SlTubulin* ([Bustin et al. 2009](#)). Each sample represents 3 biological replicates, and each biological replicate includes 3 technical replicates. The primers are listed in [Supplementary Table S5](#).

Determination of endogenous H_2S production in tomato fruit

The production of endogenous H_2S of fruit sample was determined according to the method of [Riemenschneider et al. \(2005\)](#). Fruit samples (1 g) of 35, 40, and 45 DPA were ground in liquid nitrogen and homogenized by adding 5 mL of phosphate-buffered salt (pH 7.4) solution containing 10 mM L-cysteine and 10 mM pyridoxal 5'-phosphate and then centrifuged at $10,000 \times g$ for 10 min. H_2S released from the supernatant was absorbed by zinc acetate and then reacted with N,N-dimethyl-phenylenediamine and

$FeCl_3$ to form methylene blue, then the absorbance was determined at 670 nm. The experiment was performed in 3 technical replicates and repeated for 3 biological replicates.

Yeast 2-hybrid assays

Full-length coding sequence of SIWRKY6 was cloned into the pGADT7 linearized vector and those of SIMAPK1, SIMAPK2, SIMAPK3, SIMAPK4, and SIMAPK5 were cloned and inserted into pGBKT7 vector. Then, the pGBKT7 and pGADT7 recombinant plasmids were co-transformed into yeast (*Saccharomyces cerevisiae*) strain Y2HGold cells (Clontech) for interaction screening. The synthetic dropout medium lacking Leu and Trp was used to test the transformants. Positive interactions were detected on medium lacking Leu, Trp, His and Ade ([Yao et al. 2023](#)). After the screening of interacting protein, SIWRKY6 is divided into 3 segments: amino acid region 1 to 293, 294 to 354, and 355 to 550, and SIWRKY6^{C396A}, SIWRKY6^{S33A} site mutations were constructed in pGADT7 vector, and yeast 2-hybrid were carried out between SIWRKY6 truncations, SIWRKY6 site mutations and SIMAPK4. The primer sequences are listed in [Supplementary Table S6](#).

Pull-down assays

To further explore the interaction between SIWRKY6 and SIMAPK4, CDSs of SIMAPK4 and SIWRKY6 were cloned into pGEX-4T-1 and pCold vectors to generate GST-SIMAPK4 and HIS-SIWRKY6, respectively. Recombinant GST-SIMAPK4 and HIS-SIWRKY6 were expressed from *E. coli* Rosetta (DE3) by adding 0.5 mM IPTG overnight at 16°C for 16 h. The pull-down analyses were performed referring to the method of [Miernyk and Thelen \(2008\)](#). The protein blots were analyzed using an anti-GST antibody and anti-HIS antibody (working dilution 1:5,000) from Beyotime (China). The primer sequences are listed in [Supplementary Table S3](#).

FLC assay

According to [Yao's et al. \(2023\)](#) method for FLC assay, SIWRKY6 was labeled to the N-terminal end of LUC to form NLuc-SIWRKY6, while SIMAPK4 was labeled with the C-terminal end of LUC to form CLuc-SIMAPK4. NLuc-SIWRKY6 and CLuc-SIMAPK4 constructs were then transferred into *Agrobacterium* separately, mixed at a ratio of 1:1 (v/v) and co-infiltrated into *Nicotiana benthamiana* leaves. After dark treatment for 12 to 16 h and then transferred to light incubation for 2 to 3 d, the firefly luciferase activity was measured by Enhanced Firefly Luciferase Reporter Gene Assay Kit (Beyotime, China). The primer sequences are listed in [Supplementary Table S7](#).

BiFC assay

The BiFC assay was carried out based on the method of [Song et al. \(2023\)](#). Recombinant vectors SIWRKY6-cYFP and SIMAPK4-nYFP were transfected into *Agrobacterium* strain GV3101 separately, and transient expressions were performed in *N. benthamiana* leaves of 4 wk growth. A laser confocal scanning microscope (Nikon, Japan) with an excitation wavelength at 511 nm and emission wavelength of 525 nm under laser intensity of 20% were used to observe the fluorescence signal in leaf cells after 2 d of infiltration. The primer sequences are listed in [Supplementary Table S7](#).

In vitro kinase activity assays

The in vitro kinase activity assays were performed as previously described ([Ding et al. 2018](#)). The recombinant GST-SIMAPK4 and HIS-SIWRKY6 proteins were purified using Protein Purification

Kit (Cytiva, 17075601 and 17531806) according to the manufacturer's instructions. The purified GST-SlMAPK4 and HIS-SlWRKY6 proteins were dissolved with 50 μ L of kinase reaction buffer (25 mM Tris-HCl pH 7.5, 2 mM DTT, 10 mM MgCl₂, 200 μ M ATP, protease inhibitor, and phosphatase inhibitor) and incubation for 1 h at 37°C. Dephosphorylation was carried out by the addition of 0.5 μ L Lambda Protein Phosphatase (λ -PPase). The reaction was stopped by adding 1 $\times$ SDS loading buffer and incubated at 100°C for 10 min. Samples were subsequently separated in 10% (v/v) Phos-tag SDS-PAGE gels (ApexBio, USA). The protein samples were then analyzed by immunoblotting with anti-HIS and anti-GST antibodies (working dilution 1:5,000) from Beyotime (China). Meanwhile, potential MAPK phosphorylation (SP) sites of SlWRKY6 (including S33, S229, S248, and S532) were sequentially mutated to mutants (S to A) which might be phosphorylated and cloned for protein expression and further used for phosphorylation experiments.

Dual-luciferase reporter system assay

The dual-luciferase experiment was assayed following the method described previously (Yuan et al. 2021). The gene promoter sequences of SlSGR1 and SlSAG12 upstream 1,000 bp sequence of the starting codon were inserted into the pGreenII 0800-LUC vector at the sites of HindIII and BamHI. Meanwhile, the coding sequences of SlWRKY6, mutant forms SlWRKY6^{C396A}, SlWRKY6^{S33A}, SlWRKY6^{S33D}, and SlMAPK4 were cloned into the pSAK277 vector. All recombinant constructs were transferred into *Agrobacterium* strain GV3101 including the helper PSoup and P19 plasmid. The *Agrobacteria* containing recombinant vectors were mixed with a 1:9 (v/v) ratio, and co-infiltrated into 4- to 6-wk-old *N. benthamiana* leaves. After dark treatment for 12 to 16 h and then transferred to light incubation for 2 to 3 d, total protein was extracted from the infiltration zone of *N. benthamiana* leaves. Plants of *N. benthamiana* were also fumigated with 500 μ M NaHS for 4 h to study the effect of SlWRKY6 persulfidation on transcriptional activity to SlSGR1 and SlSAG12 promoter. The ratio of firefly luciferase (LUC) and renilla luciferase (REN) was tested by using the Dual-Luciferase Reporter Assay System (E1910, Promega, USA) according to the manufacturer's instructions. LUC activity was analyzed by infiltrating *N. benthamiana* leaves with 0.5 mM D-fluorescein (Segnon, Shanghai, China) in dark conditions for 5 min. Images of leaves were acquired by an in vivo imaging system (Tanno5200, Tanon, China). The primer sequences are listed in [Supplementary Tables S4 and S8](#).

Electrophoresis mobility shift assays

EMSA were performed as described by Tan et al. (2018). The recombinant purified GST-SlMAPK4 and HIS-SlWRKY6 proteins single or HIS-SlWRKY6 treated with NaHS (500 or 1,000 μ M) for 1 h were incubated with biotin-labeled probes containing the W-box of SlSAG12 and SlSGR1 promoters ([Supplementary Table S7](#)) in binding buffer. Then the 10% of non-denaturing polyacrylamide gel was used to separate the protein-DNA complexes by 0.5 $\times$ Tris-buffered EDTA buffer (pH 8.0), and the DNA in the gels was electroblotted onto nitrocellulose membranes by Western blotting, and the chemiluminescence kit (Beyotime, China) was used to detect protein-DNA interactions. All the above primer sequences used are listed in [Supplementary Table S9](#).

VIGS and transient transformation in tomato fruits

The 400 bp fragment of SlMAPK4 was selected by the VIGS online tool (<https://vigs.solgenomics.net/>), and then the fragment was amplified by PCR and ligated into pTRV2 vector. The plasmid

was transformed into *A. tumefaciens* strain GV3101. VIGS in tomato fruits was performed as described by [Fantini and Giuliano \(2016\)](#).

Transient tomato fruit injection experiment was conducted with slight modification according to the method of [Sun et al. \(2023\)](#). The coding sequences of SlWRKY6^{C396A}, SlWRKY6^{S33A}, and SlWRKY6^{S33D} mutations were inserted into the pSAK277 vector. The plasmids were transformed into *Agrobacterium* strain GV3101 and then *Agrobacterium* was resuspended in the infection solution (10 mM MgCl₂, 10 mM 2-morpholinoethanesulphonic acid, MES; and 100 μ M acetosyringone) for transient transformation in *slwrky6* mutant fruits which were further subjected to H₂S fumigation released from 500 μ M NaHS or water control. The empty vector (pSAK277) was used as a negative control. Tomato fruit phenotypic changes were observed and sampled for further analysis. All the above primer sequences used are listed in [Supplementary Table S10](#).

Statistical analysis

All experiments use at least 3 biological replicates. One-way ANOVA was analyzed using SPSS statistical software version 24.0 (SPSS Inc., Chicago, IL, USA). Three independent experiments (replications) were used to calculate the mean and standard deviation (SD). Significant differences among the parameters means ($P < 0.05$) were analyzed by Student's t-test or Duncan's multiple tests. The data processing and graphics rendering were plotted by Origin Pro 9.0.

Accession numbers

Sequence data and accession numbers in this article were obtained by the Phytozome data libraries (<https://phytozome-next.jgi.doe.gov/>) under the following accession numbers: SlWRKY6 (Soly02g080890), SlSGR1 (Soly08g080090), SlSAG12 (Soly02g076910), SlACO1 (Soly07g049550), SlACS2 (Soly01g095080), SlMAPK1 (Soly06g005170), SlMAPK2 (Soly01g094960), SlMAPK3 (Soly05g049970), SlMAPK4 (Soly08g014420), and SlMAPK5 (Soly11g072630). The data are shown in [Supplementary Tables S1 to S10](#).

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Author contributions

K.H. and H.Z. designed the experiments. M.Z., L.M., and M.G. performed the experiments and collected the data. K.H., C.Z., G.Y., and H.Z. guided this research. M.Z. and K.H. wrote and revised the manuscript. All the authors read, revised, and approved the manuscript.

Supplementary data

The following materials are available in the online version of this article.

Supplementary Figure S1. Role of H₂S fumigation in delaying WT tomato fruit ripening.

Supplementary Figure S2. Overview of RNA-seq datasets in SlWRKY6-OE and WT tomato leaves grown for 60 d.

Supplementary Figure S3. SlWRKY6 is a positive regulator of tomato leaf senescence.

Supplementary Figure S4. Analysis of SIWRKY6, SISR1, and SISAG12 gene transcription during tomato fruit ripening.

Supplementary Figure S5. SIWRKY6 promotes the expression of ethylene synthesis genes SIACO1 and SIACS2 in tomato fruits and leaves.

Supplementary Figure S6. Analysis of tissue expression of SIMAPK1, SIMAPK2, SIMAPK3, SIMAPK4, and SIMAPK5 in tomato.

Supplementary Figure S7. Interaction between SIWRKY6 in different truncations or SIWRKY6^{C396A}, SIWRKY6^{S33A}, and SIMAPK4.

Supplementary Figure S8. Effect of SISR1 and SISAG12 gene silencing by VIGS on tomato fruit ripening.

Supplementary Table S1. Primer sequences for SIWRKY6-Cas9 and overexpression.

Supplementary Table S2. Primer sequences for identification of *slwrky6* mutants.

Supplementary Table S3. The primer sequences used for constructing prokaryotic expression vectors.

Supplementary Table S4. Primer sequences for SIWRKY6 (Cys and Ser) site mutations.

Supplementary Table S5. Primer sequences for the gene expression analysis.

Supplementary Table S6. Primer sequences for the yeast 2-hybrid assays.

Supplementary Table S7. The FLC and BiFC primer sequences.

Supplementary Table S8. The primer sequences used for dual-luciferase assays.

Supplementary Table S9. The probe sequences used for electrophoresis mobility shift assays.

Supplementary Table S10. The VIGS primer sequences.

Supplementary Data Set 1. Persulfidation site analysis data of SIWRKY6 by LC-MS/MS.

Supplementary Data Set 2. Statistics of differently expressed genes in SIWRKY6 overexpressing and WT leaves.

Supplementary Data Set 3. Differential gene expression levels in SIWRKY6 overexpressing and WT leaves.

Supplementary Data Set 4. KEGG enrichment of differential gene analysis data in SIWRKY6 overexpressing and WT leaves.

Supplementary Data Set 5. Original data of ripening-associated DEG gene heat map.

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Conflict of interest statement. None declared.

Data availability

The authors confirm that the data supporting the findings of this study are available within the article and its [supplementary materials](#).

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